Exercise 3 - Pandas and plotting with RNA-seq results

A lab grew human cells in two conditions, with 3 replicates each:

- **normoxia**: normal oxygen (samples `N1`, `N2`, `N3`)
- **hypoxia**: low oxygen, like inside a tumour (samples `H1`, `H2`, `H3`)

They sequenced the RNA of all 6 samples (RNA-seq) and ran a differential expression analysis (DESeq2) to find which genes change in hypoxia. The results came back as an **Excel file**, which is very common in bioinformatics.

Your job is to explore these results with pandas.

Beginners: use the `Exercise 2 – step by step – day 4` whenever you get stuck, and please ask if something is unclear.

Advanced: try to solve everything without the hints.

The file

The file `hypoxia_rnaseq.xlsx` has two sheets:

- **samples**: information about each sequenced sample
- **DE_results**: one row per gene, with these columns:

Column	Meaning
`gene`	gene name
`chromosome`	the chromosome of the gene
`baseMean`	average expression of the gene over all samples
`log2FoldChange`	change in hypoxia compared to normoxia, on a log2 scale: `1` = 2 times higher, `2` = 4 times higher, `−1` = 2 times lower, `0` = no change
`pvalue`	p-value of the test
`padj`	p-value adjusted for testing many genes. **Use this one**: `padj < 0.05` means *significant*

Missing values are shown as `NaN` ("not a number"). It is pandas' way of saying *no value*, similar to `None`.

Run this cell first:

In [ ]:
```
import pandas as pd

file = '../../downloads/hypoxia_rnaseq.xlsx' # relative path of the file. It
```

Part 1 - The samples

1. Read the sheet **samples** into a dataframe called `samples`, and look at it.

   `pd.read_excel()` reads the **first** sheet by default. To choose a sheet, use the keyword argument `sheet_name`.

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2. The numbers in `total_reads` are hard to read. Create a new column `million_reads` with the number of reads in millions.

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3. Make a **bar plot** with one bar per sample, showing `million_reads`. Add a title and axis labels.

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4. One sample looks different from the others. Which one? Also look at the column `percent_mapped`. What could have happened to this sample?

Your answer:

5. Calculate the **mean** number of million reads **per condition**.

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Part 2 - The results

6. Read the sheet **DE_results** into a dataframe called `results`.
   - How many genes are in the table?
   - Use `.info()`: which columns have missing values?

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7. `VEGFA` is a famous gene that cells switch on when they lack oxygen. Look at its row. Is it higher or lower in hypoxia? **How many times** higher or lower?

   Remember: the fold change is `2 ** log2FoldChange`.

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8. Select the **significant** genes (`padj` smaller than `0.05`) and save them in a variable called `significant`. How many are there?

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9. Biologists usually only care about genes that are significant **and** change at least 2 times (`log2FoldChange` larger than `1` or smaller than `-1`).

   - Select the **up-regulated** genes: `padj < 0.05` and `log2FoldChange > 1`. Save them as `up`.
   - Select the **down-regulated** genes: `padj < 0.05` and `log2FoldChange < -1`. Save them as `down`.
   - How many genes are in each?

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10. Show the **10 genes** that go up the most in hypoxia. Show only the columns `gene`, `log2FoldChange` and `padj`.

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11. Make a **histogram** of `log2FoldChange` for **all** genes, with 30 bins. Add a title and an x label. What do most genes do?

In [ ]: